



New satellite technology helped explain how Earth's magnetic field channels energetic solar particles into a collision course with the atmosphere to produce the aurora borealis, or Northern Lights.

THE SOVIET UNION'S SPUTNIK 1

—the first artificial satellite—burned up on January 4, after spending three months in orbit. The US joined the space race with the **launch of Explorer 1** later that month. For the month or so of their battery-powered lives, these satellites sent back important information about airspace—a term that refers to the realm of the atmosphere and the accessible space above it.

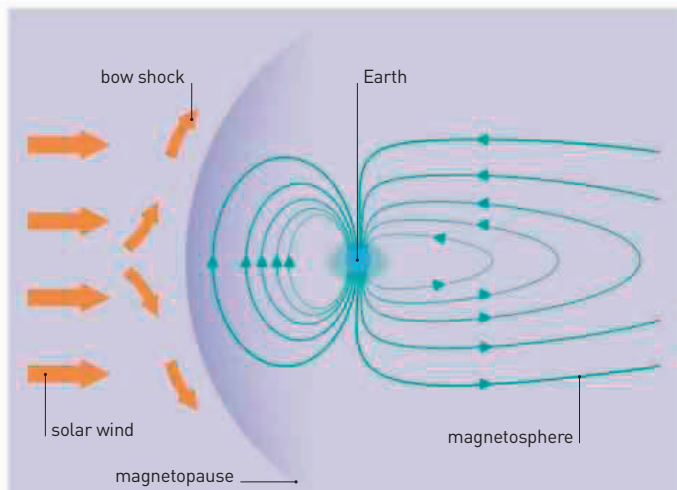
Sputnik 1 measured the **density of the upper atmosphere**, and Explorer 1 found how Earth's **magnetosphere** (see panel, below) could **deflect harmful radiation** before it reached the ground.

In California, **Matthew Meselson** (b.1930) and **Franklin Stahl** (b.1929) were unlocking the secrets of DNA in what would come to be described as “the most beautiful experiment in

biology.” They traced what happened to components of DNA when it replicated, and found that a **double helix molecule** unravelled into two strands, each of which was a **genetic “template” for making more DNA**. After replication, each new double helix contained one old strand and one new one. This method of semiconservative replication had been proposed by Watson and Crick in 1953, when they developed the double helix model of DNA.

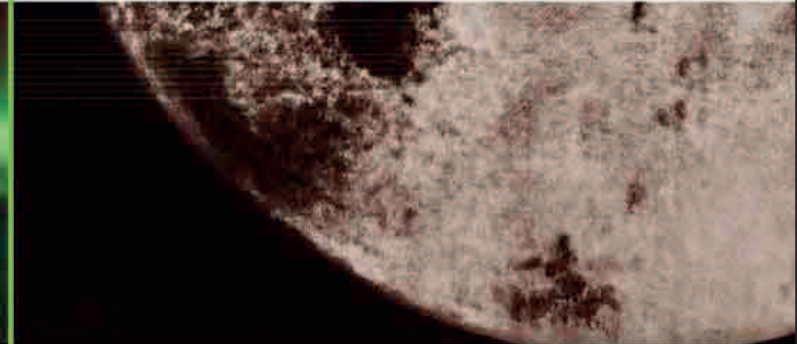
In the same year, a practical expression of this replication came with the work of British botanists **Frederick Steward** (1904–93) and **John Gurdon** (b.1933). Both teased cells from mature organisms—Gurdon from a tadpole and Steward from a carrot—and grew clones from them as new genetically identical plants. It was the **first cloning using material** from differentiated “**body**” tissues.

The year also saw a breakthrough in electronic engineering. American electrical engineers **Jack Kilby** (1923–2005) and **Robert Noyce** (1927–90) simultaneously came up with the idea of condensing all the components of an electronic circuit into a single plate of silicon, thereby inventing the **microchip**.



THE MAGNETOSPHERE

Earth is enveloped by the magnetosphere, a magnetic “blanket” that results from magnetism deep inside the planet. The magnetosphere deflects harmful high-energy particles from the Sun through a bow shock, or shock wave created at the magnetopause—an abrupt boundary between incoming solar wind and the magnetosphere. Without the bow shock, the solar wind would destroy life on Earth.



Hidden from the gaze of astronomers since antiquity, the far side of the Moon was finally revealed when USSR's Luna 3 probe photographed it from orbit.

TWO MONKEYS, ABLE AND BAKER

went up in the US missile **Jupiter AM-18**, and became the first primates to survive a space flight. Meanwhile, the **USSR** launched three **lunar probes**—Luna 1 and 3 achieved “flyby.” Luna 3 also took the first photograph of the Moon's far side.

In Cambridge University, UK, molecular biologist **Max Perutz** (1914–2002) was studying

hemoglobin—the red oxygen-carrying pigment of blood—using techniques that had cracked the structure of DNA. He found that it had four protein chains, each with oxygen-grabbing iron.

Australopithecine skull

Dated at 1.75 million years old, this skull belongs to an australopithecine, also known as “nutcracker man” because of his large cheek teeth.

